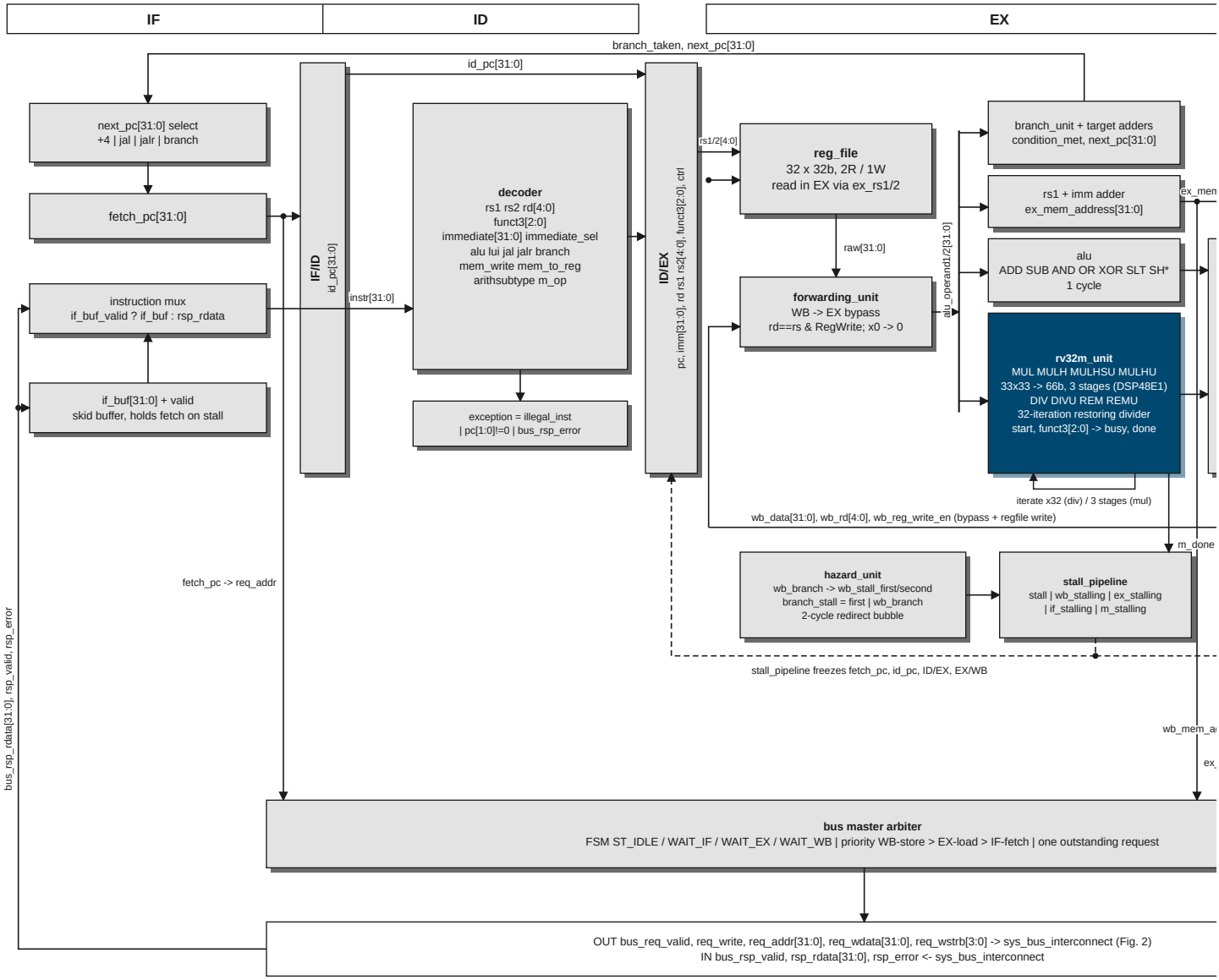
Figure 1 - RV32IM core: datapath, control and bus master

cpu_core.v | 3-stage pipeline: IF+ID | EX | MEM+WB | single clk (100 MHz target), async active-low reset on every register



Key control signals (assignment name -> RTL name)

RegWrite	wb_reg_write_en = alu|lui|jal|jalr|load|m	**ALUOp**	ex_alu_op[2:0] + ex_arithsubtype
MemRead	ex_mem_to_reg	**Branch/Jump**	ex_branch, ex_jal, ex_jalr -> branch_taken
MemWrite	ex_mem_write -> wb_mem_write	**M select**	ex_m -> m_start; m_stalling = ex_m & !m_done
ALUSrc	ex_imm_sel	**Stall**	stall_pipeline

Legend

existing RTL in rtl/cpu/core

new in this project

Honest stage mapping: IF+ID fetch and decode in EX reads regfile, executes, issues loads; MEM+W writes regfile. Dashed = stall_pipeline hold.